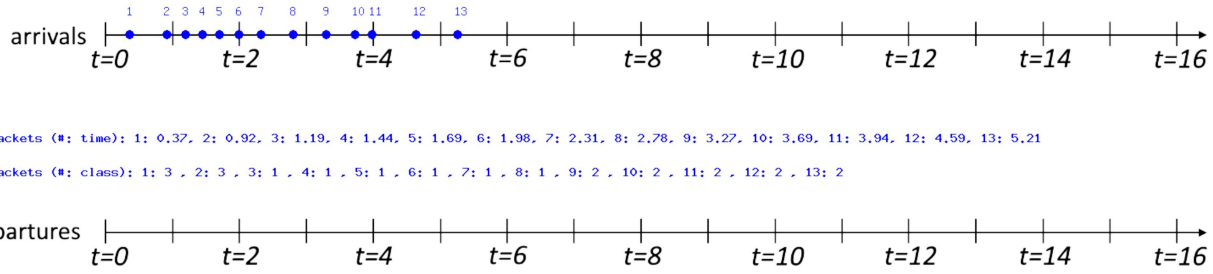
INTERACTIVE END-OF-CHAPTER EXERCISES

PACKET SCHEDULING

Consider the arrival of 13 packets to an output link at a router in the interval of time [0, 5], as indicated by the figure below. We'll consider time to be "slotted", with a slot beginning at t = 0, 1, 2, 3, etc. Packets can arrive at any time during a slot, and multiple packets can arrive during a slot. At the beginning of each time slot, the packet scheduler will choose one packet, among those queued (if any), for transmission according to the packet scheduling discipline (that you will select below). Each packet requires exactly one slot time to transmit, and so a packet selected for transmission at time t, will complete its transmission at t+1, at which time another packet will be selected for transmission, among those queued. You might want to review section 4.2.5 in the 8th edition of our textbook, on packet scheduling.

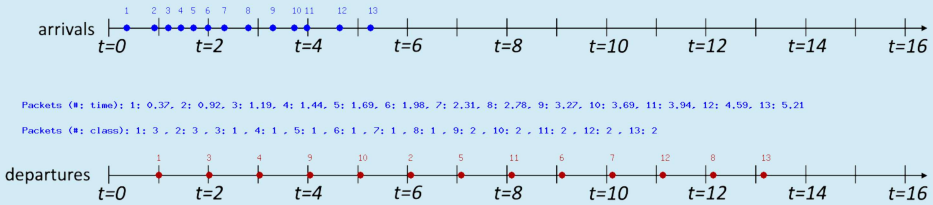


Choose a specific packet scheduling discipline (FCFS, Priority, RR, and WFQ) from the list below. In the case of Priority, RR, and WFQ there will be three classes of traffic (1, 2, 3), with lower class numbers having higher priority in the case of priority schedule, or beginning earlier in the case of RR and WFQ. In the case of WFQ, scheduling weights are 0.5, 0.3, and 0.2.

QUESTION LIST

1. At t=1, which packet is sent out? Give the packet # or 'n/a' if applicable
2. At t=2, which packet is sent out? Give the packet # or 'n/a' if applicable
3. At t=3, which packet is sent out? Give the packet # or 'n/a' if applicable
4. At t=4, which packet is sent out? Give the packet # or 'n/a' if applicable
5. At t=5, which packet is sent out? Give the packet # or 'n/a' if applicable
6. At t=6, which packet is sent out? Give the packet # or 'n/a' if applicable
7. At t=7, which packet is sent out? Give the packet # or 'n/a' if applicable
8. At t=8, which packet is sent out? Give the packet # or 'n/a' if applicable
9. At t=9, which packet is sent out? Give the packet # or 'n/a' if applicable
10. At t=10, which packet is sent out? Give the packet # or 'n/a' if applicable
11. At t=11, which packet is sent out? Give the packet # or 'n/a' if applicable
12. At t=12, which packet is sent out? Give the packet # or 'n/a' if applicable
13. At t=13, which packet is sent out? Give the packet # or 'n/a' if applicable

SOLUTION



Packets (#: time): 1: 0.37, 2: 0.92, 3: 1.19, 4: 1.44, 5: 1.69, 6: 1.98, 7: 2.31, 8: 2.78, 9: 3.27, 10: 3.69, 11: 3.94, 12: 4.59, 13: 5.21

Packets (#: class): 1: 3 , 2: 3 , 3: 1 , 4: 1 , 5: 1 , 6: 1 , 7: 1 , 8: 1 , 9: 2 , 10: 2 , 11: 2 , 12: 2 , 13: 2

1. At time t=1, the packet sent is 1.

2. At time t=2, the packet sent is 3.

3. At time t=3, the packet sent is 4.

4. At time t=4, the packet sent is 9.

5. At time t=5, the packet sent is 10.

6. At time t=6, the packet sent is 2.

7. At time t=7, the packet sent is 5.

8. At time t=8, the packet sent is 11.

9. At time t=9, the packet sent is 6.

10. At time t=10, the packet sent is 7.

11. At time t=11, the packet sent is 12.

12. At time t=12, the packet sent is 8.

13. At time t=13, the packet sent is 13.

QUESTION 1 OF 13

At t=1, which packet is sent out? Give the packet # or 'n/a' if applicable

Answer

CHECK HINT SKIP

CHANGE SCHEDULING

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

Copyright © 2010-2022 J.F. Kurose, K.W. Ross
Comments welcome and appreciated: kurose@cs.umass.edu